

A network is a diagram consisting of a number of points called nodes or vertices which are connected by line segments called arcs. The arcs may be straight or curved and must always begin and end at a node. Project networks involve completing a specified project ensuring that all tasks are completed and in a specified order.



A project is a situation which requires specified tasks to be completed in a specified order. Some of the tasks may be done simultaneously as others and other tasks may not be commenced until certain specified tasks have been completed.

A project network involves showing all the tasks of a project that need to be completed on a **directed graph** or **digraph**.

The arcs of the digraph show the order in which the tasks are to be completed and the nodes indicate the least time required to reach that stage in the project with all tasks on the paths which finish at that node having been completed.

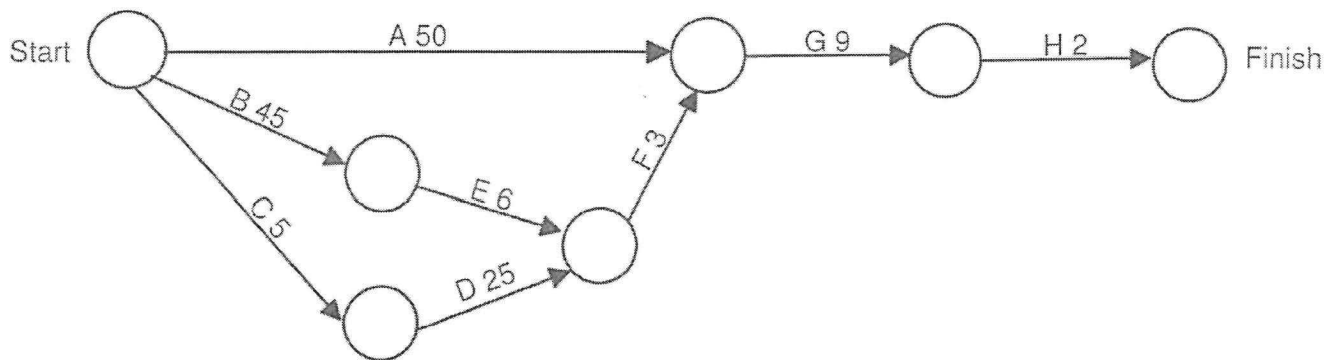
Example 1:

Consider making and baking a pizza in a wood-fired oven. The tasks that need to be completed and the times taken to do so are listed below.

TASKS	TIME TAKEN (mins)	ORDER OF TASKS
A: Heating oven	50	A, B and C can all start at the same time.
B: Making dough	45	
C: Washing vegetables	5	
D: Preparing toppings	25	D must follow C
E: Rolling dough for base	6	E must follow B
F: Placing toppings on base	3	F must follow B, C, D and E
G: Cooking pizza	9	G must follow A, B, C, D, E and F
H: Slicing and serving	2	H must follow all tasks.



The tasks and the time to complete each of them have been mapped below as a digraph, indicating the order in which they are to be completed.



Note that each node represents a stage of completion of all prior tasks on the paths which finish at that node.

We are required to find the minimum time for the completion of this project. Completing this project requires all tasks to be done, therefore the minimum completion time will be the time taken on the longest path which is called the critical path.

⇒ If any of the tasks on the critical path are delayed, then the minimum completion time will be extended by the time of the delay.

To find the minimum completion time, the critical path, and to identify any slackness we need to use a **systematic approach**. This will involve two stages as outlined below.

- Stage 1: Forward scanning
- Stage 2: Backward scanning

* Stage 1 Forward scanning:

Work from the start to the finish of the network in the direction of the arrows and in the top half of each node record the earliest starting time (EST) for every activity in the network. The EST is the least time taken to reach this stage in the project with all previous tasks in the same path completed.

* Stage 2 Backward scanning:

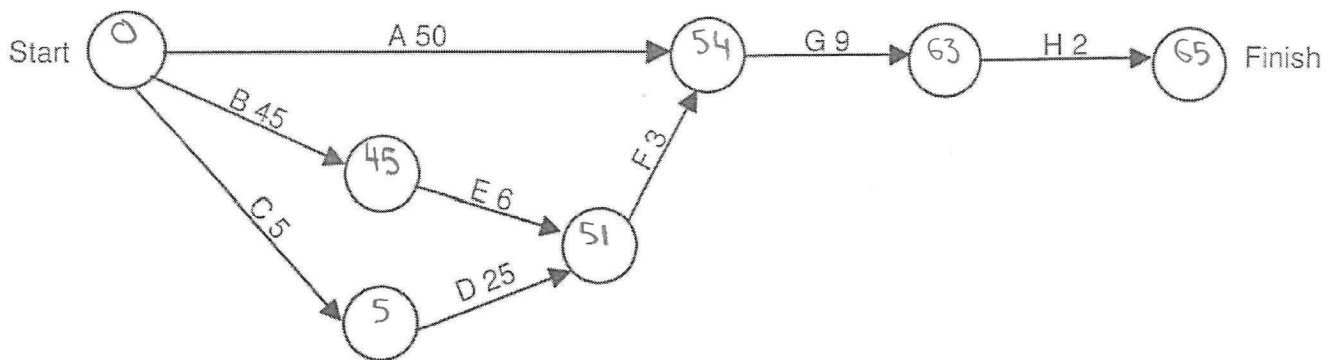
To identify any the critical path and slackness in the system begin at the finish and work back through the network recording in the lower half of each node the latest starting time (LST) for every activity in the network. The LST is the latest time that stage can be reached while still completing the project in the minimum completion time.

Stage 1 Forward scanning:

Let us apply this systematic method to the example under discussion.

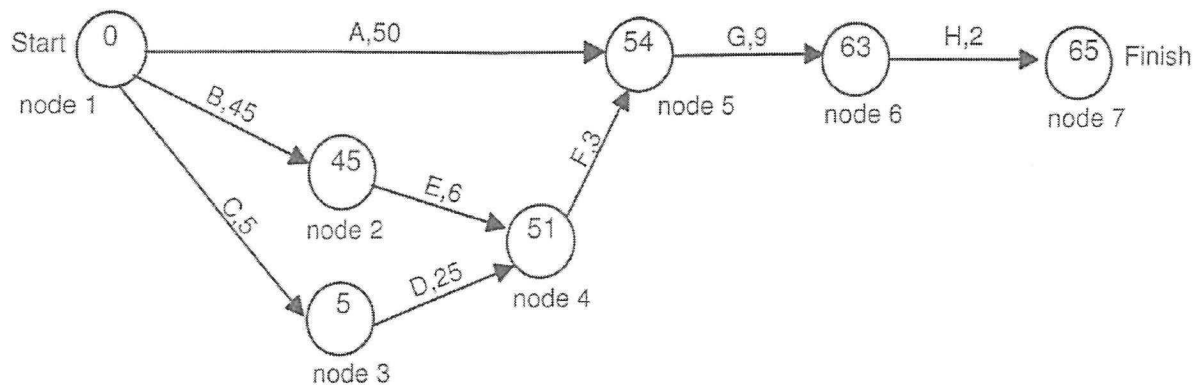
Applying stage 1, forward scanning, to the project network gives the following result.

"¹¹ Finding the largest number "11"



Hence, the minimum completion time is 65 minutes.

Solution



Notes:

- 0 is written in node 1 to indicate the start of the project and that the earliest starting time of tasks A, B and C which can all start at the same time.
- 45 is written in node 2 to indicate that task B has been completed and that the EST of task E is 45 minutes into the project.
- 5 is written in node 3 to indicate that task C has been completed and that the EST of task D is 5 minutes into the project.
- ✱ • 51 is written in node 4 because tasks B, C, D and E must all be completed and the EST of task F is 51 minutes into the project.
- 54 is written in node 5 because tasks A, B, C, D, E and F must all be completed before task G can start. That is the EST of task G is 54 minutes into the project.
- 63 is written in node 6 to indicate that task G and all preceding tasks have been completed. The EST of task H is 63 minutes into the project.
- node 7 gives the minimum completion time which in this situation is 65 minutes.

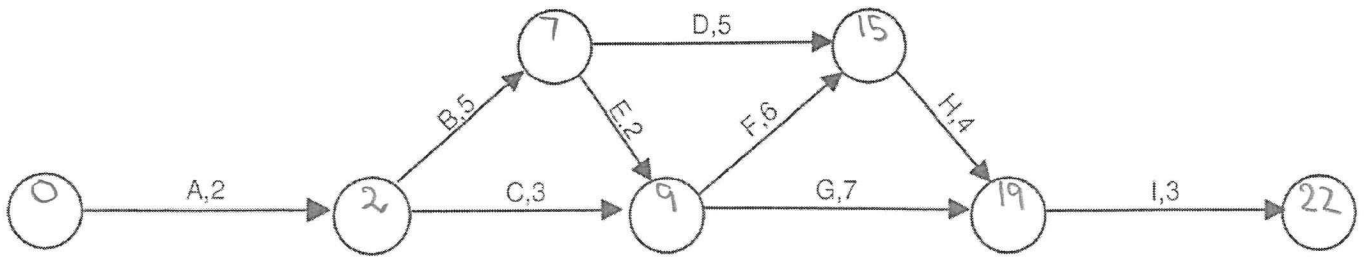
Determining the Earliest Starting Times

- At the starting node enter a 0 in the top half of the node or vertex.
- The earliest starting time (EST) for any task is the maximum time taken to reach that task. Therefore when two or more arcs or paths join a node we choose the highest total time as the EST.
- Enter the EST in every node.
- The entry in the finishing node will give the minimum completion time for the project under consideration.

Ex 1. [Ex 7A, Q1]

Complete the EST (Earliest Starting Times) for the following network.

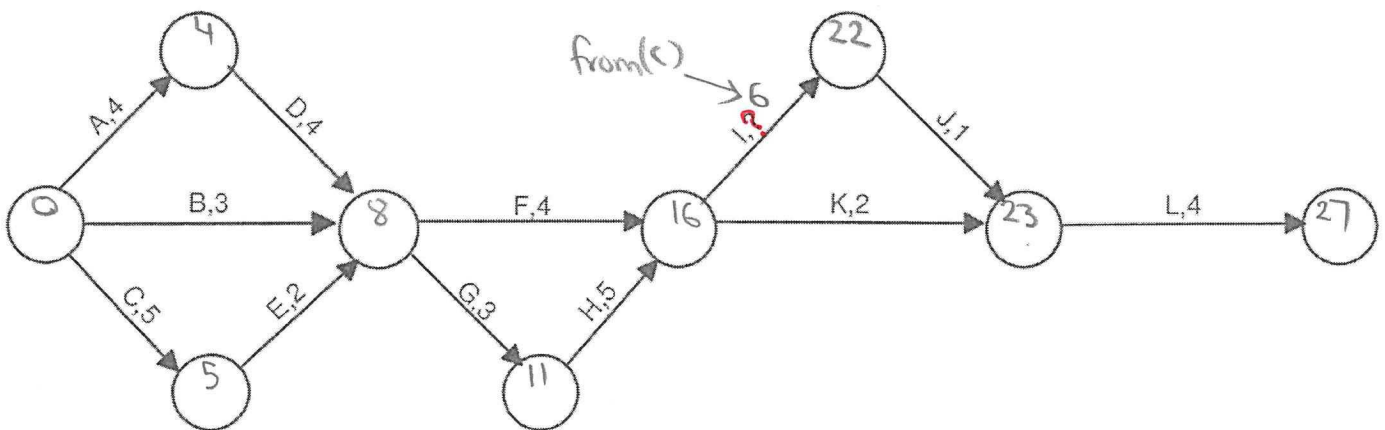
"Find the largest No."



∴ The minimum completion time is 22 units.

Ex 2. [Ex 7A, Q6]

Consider the network below with the duration of tasks A to L given in minutes.



(a) What is the EST of task G?

8 minutes

(b) What is the EST of task K?

16 minutes

(c) What is the duration of task I if the EST of task J is 22 minutes?

$22 - 16 = 6$ minutes
Task I = 6 minutes

(d) What is the minimum completion time of this project?

The minimum completion time is 27 minutes.

Complete Ex 7A

✖

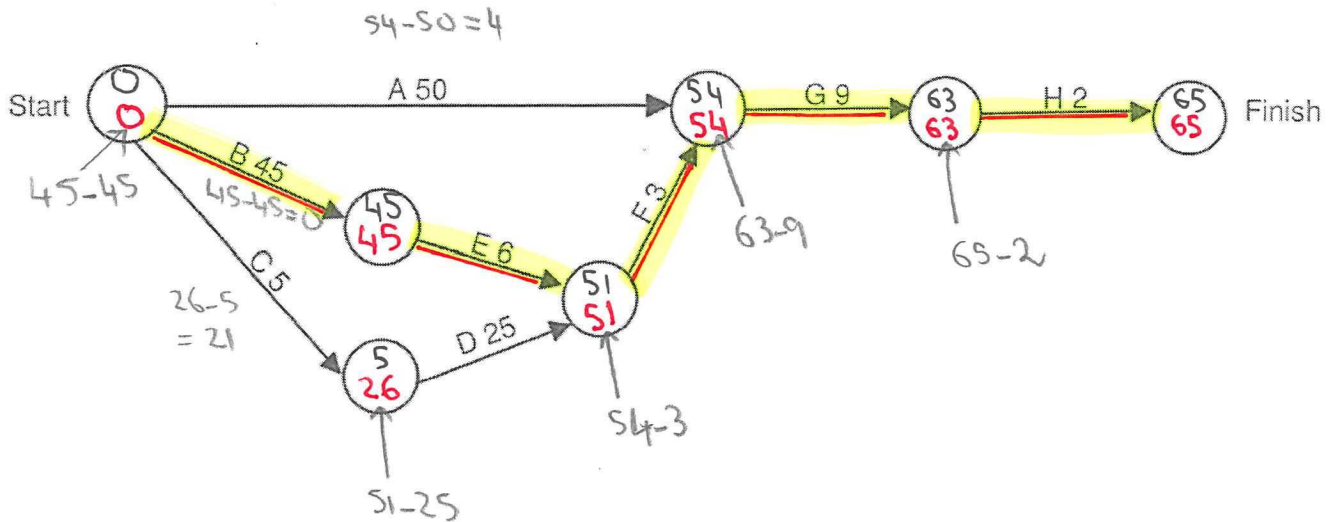
Stage 2 Backward scanning:

Applying stage 2, backward scanning, will enable us to identify the critical path and if there is any slack and thus the amount of any slackness in the network system while still completing the project in the determined minimum completion time.

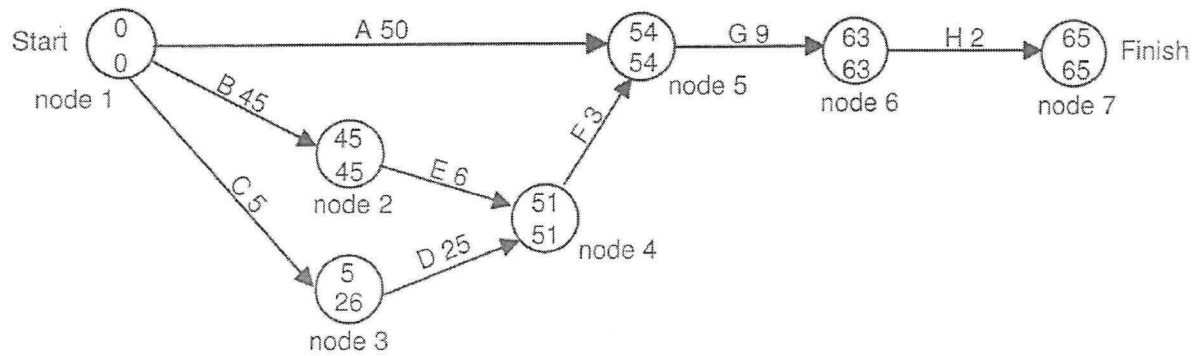
Backward scanning through the network enables us to find the **latest starting time (LST)** for each of the tasks. The LST is the latest time we can start a task without affecting the minimum completion time of the project.

⇒ "Find the smallest number"

From Example 1:



Solution

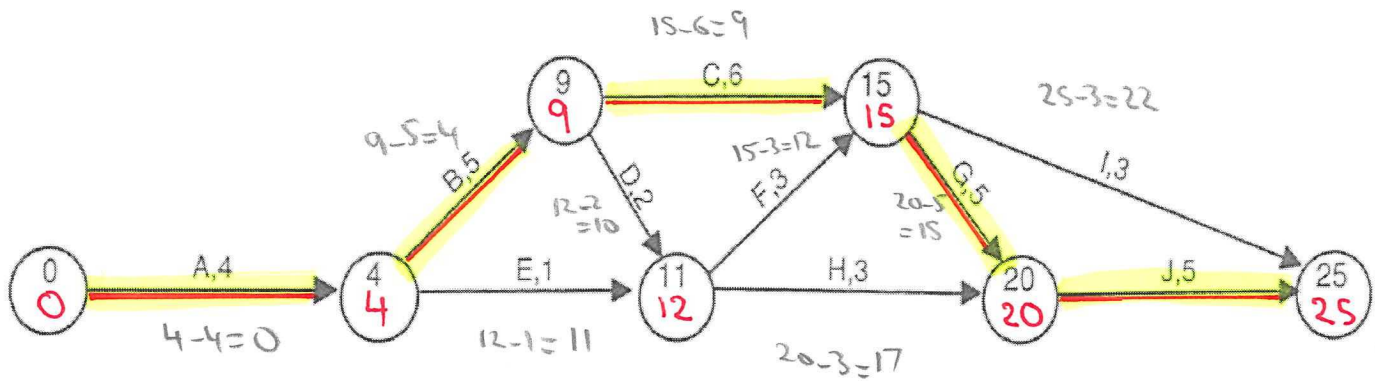


Determining the Latest Starting Times

- (i) Start at the finishing node, that is node 7, and enter, in the lower half of the node, the earliest starting time for that node as it is the same as the latest starting time.
- (ii) 63 is entered in the lower half of node 6 and has been determined by subtracting the task time of task H from the latest starting time found in node 7.
- (iii) 54 is entered in the lower half of node 5 and has been determined by subtracting the task time of task G from the latest starting time found in node 6 and so on.
- (iv) Nodes 2, 3 and 5 backtrack to node 1 and when there is more than one path to the previous node the smallest result is taken as the LST. In this case backtracking along task C gives an LST of 21, backtracking along task B gives an LST of 0 and backtracking along task A gives an LST of 4. Now 0 is the smallest LST and is entered in the lower half of node 1.

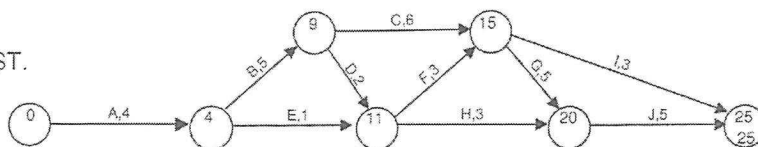
Ex 3.

For the project network shown below the earliest starting times (EST), in hours, have been entered. Complete the node entries by inserting the latest starting times for each of the activities.



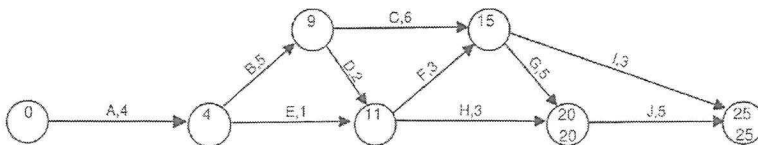
SOLUTION

In the last node, enter 25 in the lower half, as the LST is the same as the EST.



As there is only one path backwards to the node at the start of activity J, the LST = $25 - 5 = 20$.

Hence 20 is entered in the lower half of the node at the start of activity J.

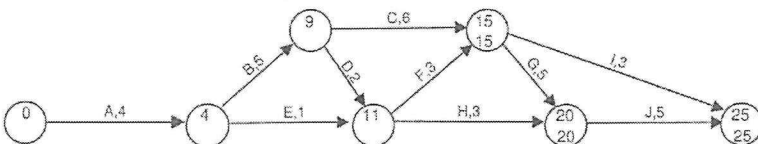


There are two paths backwards to the node at the start of activities I and G.

LST for activity I = $25 - 3 = 22$

LST for activity G = $20 - 5 = 15$

The smaller is 15 and is entered in the lower half of the node at the start of activities I and G.

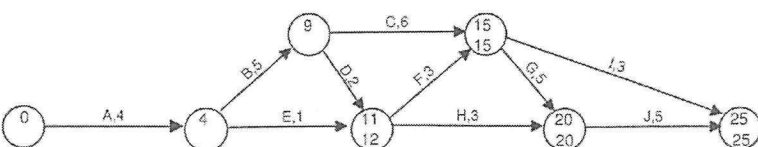


There are two paths backwards to the node at the start of activities H and F.

LST for activity H = $20 - 3 = 17$

LST for activity F = $15 - 3 = 12$

The smaller is 12 and is entered in the node at the start of activities H and F.

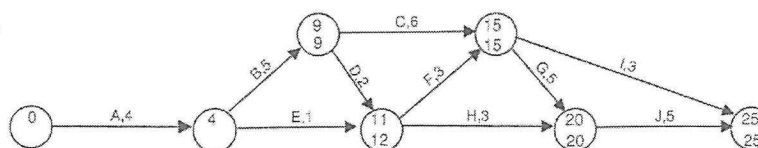


There are two paths backwards to the node at the start of activities D and C.

LST for activity D = $12 - 2 = 10$

LST for activity F = $15 - 6 = 9$

The smaller is 9 and is entered in the node at the start of activities D and C.

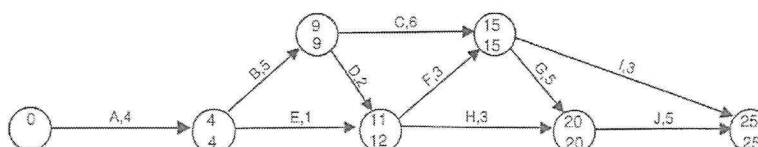


There are two paths backwards to the node at the start of activities E and B.

LST for activity E = $12 - 1 = 11$

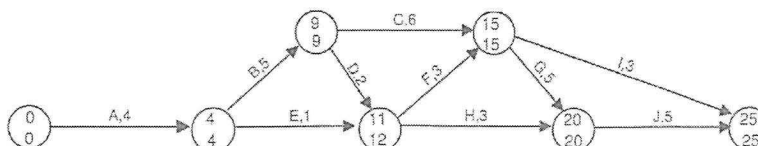
LST for activity F = $9 - 5 = 4$

The smaller is 4 and is entered in the node at the start of activities E and B.



As there is only one path backwards to the node at the start of activity A, the LST = $4 - 4 = 0$.

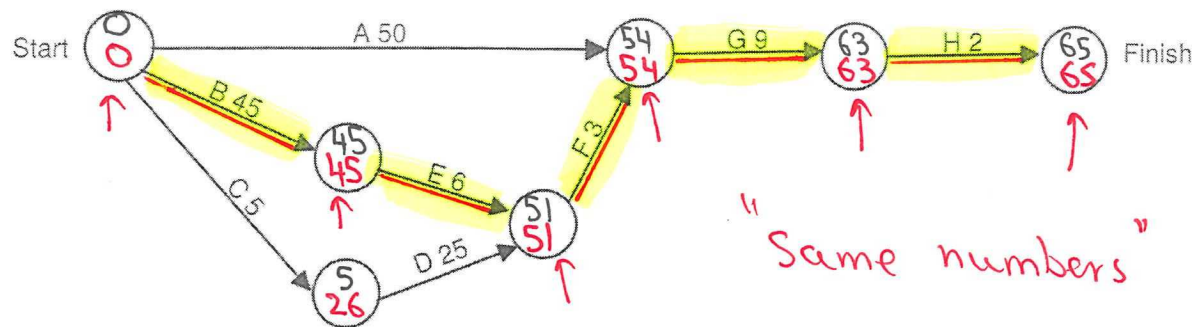
Hence 0 is entered in the starting node.



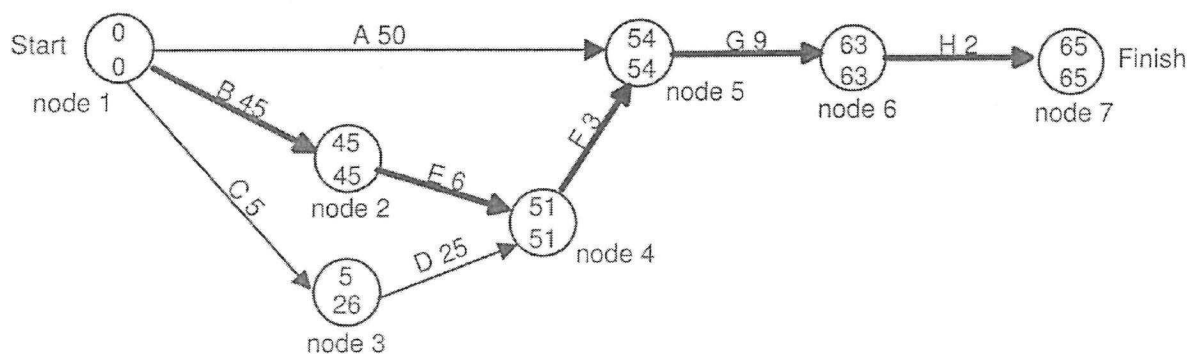
Finding the Critical Path

The critical path for a project network is the longest path through the network and hence the minimum time in which the project can be completed.

- * To determine the critical path we can calculate the length, in terms of time taken, of all the paths from the start to the finish of the project.



To find the critical path through a project network we make use of the earliest starting time (EST) and latest starting time (LST) values. The network below shows the EST and LST values for this project.



Note that the nodes of the critical path B-E-F-G-H contain EST and LST entries which are identical. This enables us to identify:

- (i) the critical path, and
- (ii) that there is no slack at these particular points of the project.

Nodes that are not on the critical path have entries which are not identical and indicate that there is slack (or float) at these points.

- * NOTE: Project networks may have more than one critical path.

Float or Slack

Float or slack is the maximum amount of time that a task or activity in a project network can be delayed without affecting the completion time of the project.

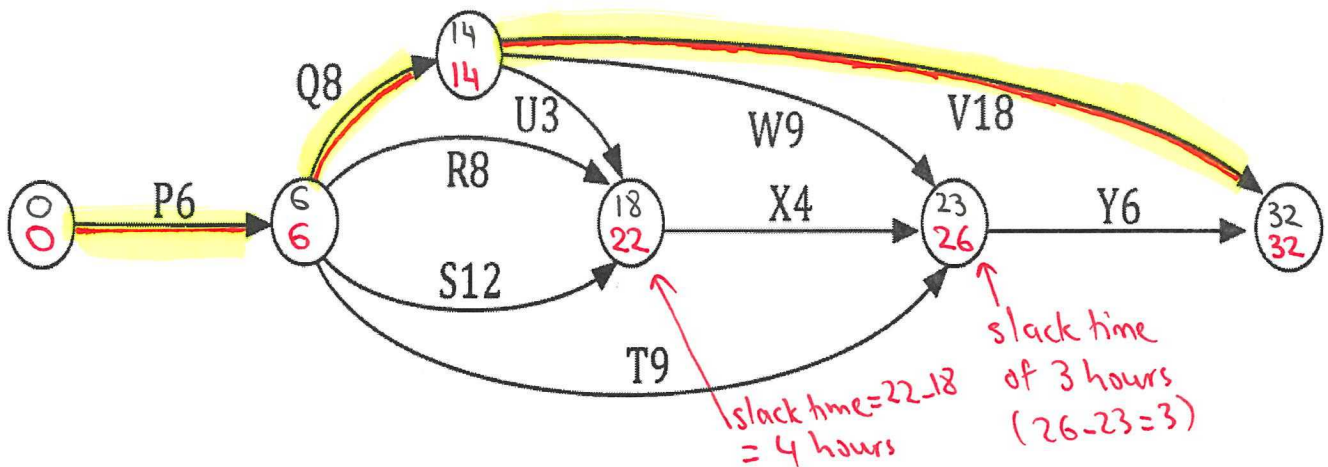
All tasks or activities on the critical path are called **critical activities** or tasks and have a float of zero.

All tasks or activities not on the critical path are called **non-critical activities** or tasks and will have a float of more than zero.

✗ ✗ To determine the amount of slack or float for a particular activity we can use the formula:
Float, for say, activity X = LST at the end of X – EST of X – duration of X

Ex 4.

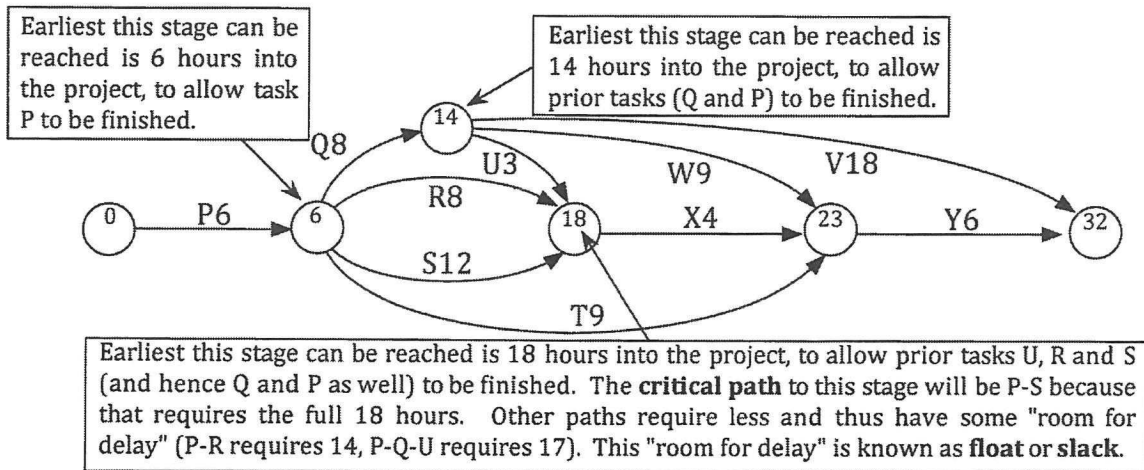
Find the **minimum completion time** and the **critical path** for the project network shown below. The arcs P, Q, ... , Y represent the tasks involved and the numbers on the arcs indicate the time in hours that the task requires.



What is the maximum number of extra hours that could be allocated to task S, over and above the 12 in the network, without extending the minimum completion time?

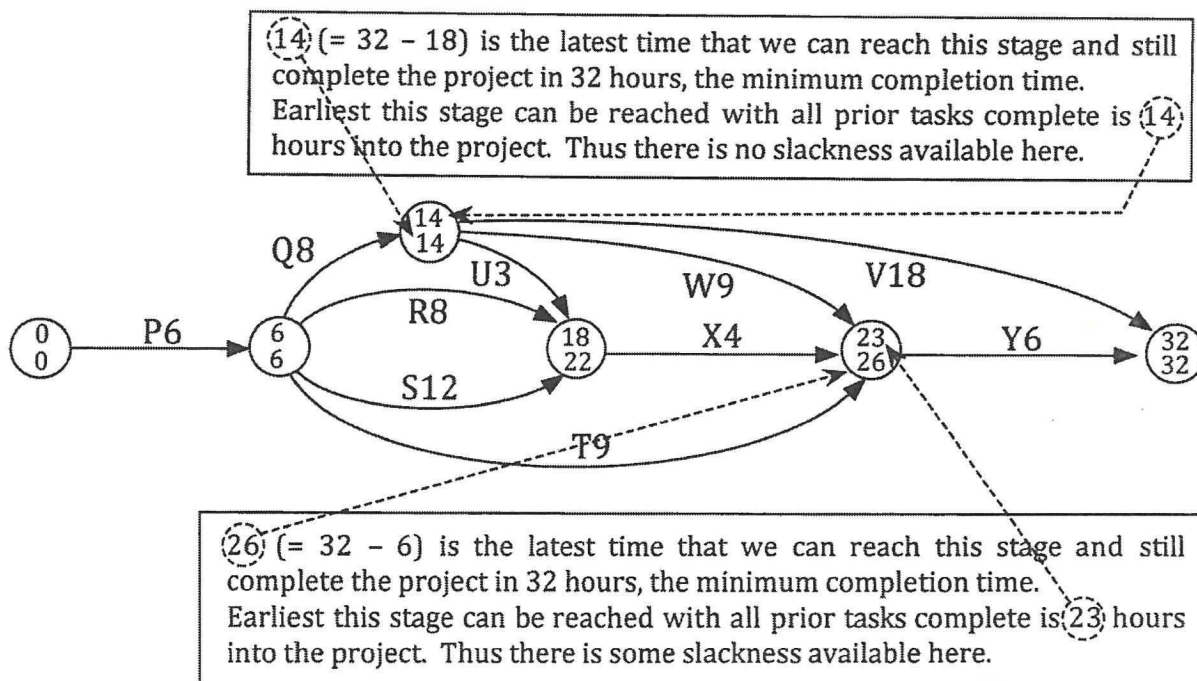
- The minimum completion time is 32 hours.
- The critical path is P-Q-V.
- Task S:
 $\Rightarrow$ slack time of 4 hours ($22 - 18 = 4$ hours)

Solution



Thus the **minimum completion time** is 32 hours.

Now, to identify the slackness in the system, work back through the network and at each vertex record in the lower half of the circle the latest time we could reach that stage whilst still being able to complete the project in the minimum completion time. Check that you understand the placement of these numbers in the bottom halves of the circles below.



Any vertex for which the earliest time the vertex can be reached is the same as the latest time the vertex can be reached lies on the **critical path**. If we journey from start to finish, through such vertices we can obtain the critical path, P - Q - V. If any of these tasks are delayed the minimum completion time will be extended.

Consider task S and notice that this task will be ready to start 6 hours into the project and does not have to finish until 22 hours into the project. Thus we could allow 16 hours for this 12 hour task. Thus we could allocate an extra 4 hours for task S, over and above the 12 in the network. Task S is said to have a **float**, or **slack** time of 4 hours.

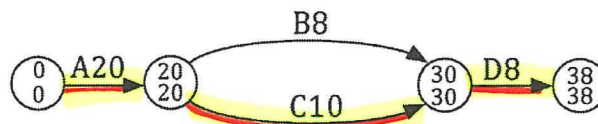
Consider task X. We say that X has an **earliest start time** (EST) of 18 hours and a **latest start time** (LST) of 22 hours ($26 - 4$). X has 4 hours float.

Consider task W. It has an **earliest start time** (EST) of 14 hours and a **latest start time** (LST) of 17 hours ($26 - 9$). W has 3 hours float.

*** * All activities on the critical path have zero float time.**

- Note:
- In the above example the diagram is drawn twice and comments are included in boxes. The reader would not have to include such comments and would only need one diagram.
 - The float, or slack, considered here is the time by which an activity can be delayed, from its early start time, without altering the project's minimum completion time. It is sometimes referred to as the total float of an activity.

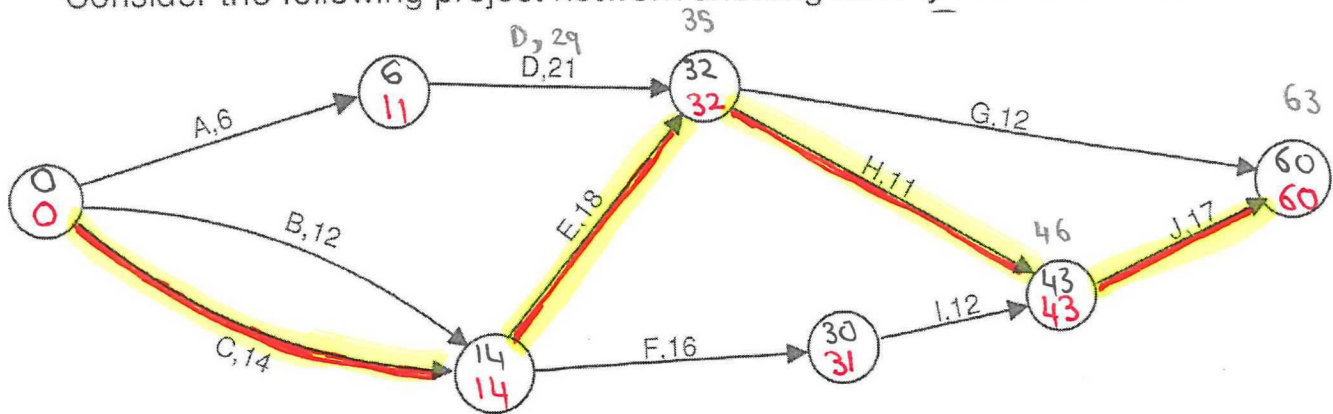
- * • Consider the diagram:**



The numbers in the circles may at first glance give the impression that there is no float, or slack, in the system but with two arcs from the second vertex to the third vertex the critical path is determined by taking task C, the task that takes the greater time. Thus A - C - D is the critical path. Task B has 2 units of float, a fact that can be missed if we only look at the numbers in the circles.

Ex 5.

Consider the following project network showing activity duration in hours.



- How many activities are required to be completed for this project?
- State the critical path and the minimum completion time for the project.
- List the critical activities and determine their float
- Determine the float times for all non-critical activities.
- If activity D has a float of 5 hours, what does this mean?
- What is meant by the term 'slack'?
- If activity J could be completed in 10 hours, how would the critical path and minimum completion time be affected?

This project commenced at 8:30 am on Monday.

- What would be the latest starting time for activity F?
- If activity D was delayed by 8 hours, what would be the new completion time?
- How many people will be required to complete this project in the minimum completion time?

a) Ten activities: A, B, C, D, E, F, G, H, I, J.

b) The critical path: C-E-H-J.

The minimum completion time is 60 hours.

c) The critical activities are C, E, H, J. Each of these activities is on the critical path, so each has a float (slack) = 0.

d) A: float = $11 - 6 = 5$ hours

B: float = $14 - 12 = 2$ hours

D: float = $11 - 6 = 5$ hours
[OR = $32 - 21 - 6$]

F: float = $31 - 30 = 1$ hour.

G: float = $60 - 12 - 32 = 16$ hours

I: float = $43 - 12 - 30 = 1$ hour.

e) Activity D has a float of 5 hours means that the starting time of D can be delayed by up to 5 hours without affecting the minimum completion time of the project. In other words, Activity D can have 26 hours ($21 + 5$) instead of 21 hours.

f) slack or float is the amount of time an activity can be delayed without affecting the minimum completion time.

g) The critical path is the same but the minimum completion time is decreased by 7 hours to 53 hours ($60 - 7$).

h) The EST (Earliest Starting Time) for F is 14 hours plus 1 hour of float time:

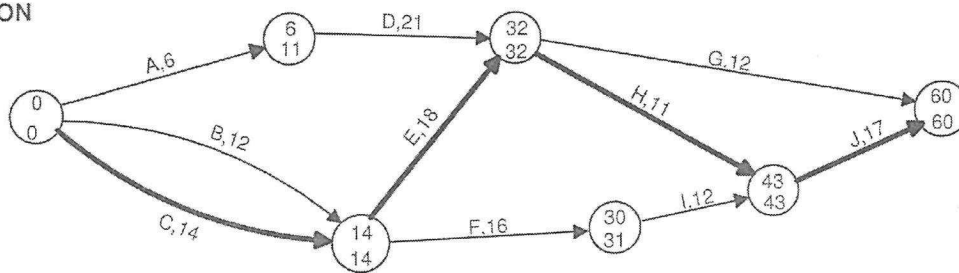
⇒ It can start 15 hours ($14 + 1$) into the project.

⇒ The LST (latest start time) = 8:30 am + 15 = 23:30 or 11:30 pm.

i) The new completion time is 63 hours.

j) 3 people { Person 1: C, E, H, J
Person 2: A, D, G
Person 3: B, F, I } combinations are possible.

SOLUTION



- Ten activities, labelled A, B, C through to J.
- Critical path is C-E-H-J. Minimum completion time is 60 hours
- Critical activities are C, E, H and J. As each of these activities is on the critical path each has a float of zero hours.
- Activity A float = $11 - 0 - 6 = 5$ hours

Activity B float = $14 - 0 - 12 = 2$ hours

Activity D float = $32 - 6 - 21 = 5$ hours

Activity F float = $31 - 14 - 16 = 1$ hour

Activity G float = $60 - 32 - 12 = 16$ hours

Activity I float = $43 - 30 - 12 = 1$ hour
- A float of 5 hours for activity D means that the starting time of D can be delayed by up to 5 hours without affecting the minimum completion time of the project. Or the time taken to complete activity D can be a total of $21 + 5 = 26$ hours instead of 21 hours.
- Slack is another term for float and it means the amount of time an activity can be delayed without affecting the minimum completion time.
- Critical path not affected, minimum completion time decreased by 7 hour to 53 hours.
NOTE: We have reduced the completion time of the project from 60 hours to 53 hours by employing more resources in carrying out activity J. Activity J is a critical activity as it lies in the critical path. When we reduce the completion time of a critical activity the process is called **crashing**.
- The earliest starting time for F is 14 hours into the project, now activity F has 1 hour of float time and hence it can start 15 hours after the project start time. Therefore the latest start time for F is Monday 11:30 pm.
- New completion time is 63 hours.
- 3 people. Person 1 to do the activities on the critical path, person 2 to do activities B, F and I, person 3 to do activities A, D and G. (other combinations of non-critical activities for persons 2 and 3 are possible)